Social dynamics, explained from the bottom up

Marion Moranetz · Los Altos Hills · Paper v1.0 · ultimate causes of social behaviour

CITATION GATE

Gate run 2026-09-04, and it is a *partial* gate: this paper was written before the series adopted per-claim grading, and its citations are not tagged inline. What was done is stated exactly. Eight load-bearing sources were checked against a primary source this pass and are listed as verified in the references. The remainder are foundational works in evolutionary biology and social psychology, listed with full details but **not individually re-confirmed**. Treat an untagged citation here as canonical-but-unverified rather than as carrying the same warrant as a VERIFIED tag elsewhere in the series.

ABSTRACT

Accounts of social behaviour usually explain a move by its immediate motive — politeness, fear, liking, ambition. This paper argues those are proximate descriptions of a single underlying regulatory system. Because human survival was obligately outsourced to the group, selection built organisms whose master computational task is the continuous estimation, signalling, and defence of their *value to the collective* — the quantity that, ancestrally, determined whether one was fed, mated, protected, or expelled. On this account "social dynamics" is not a grab-bag of phenomena but the traffic of one distributed value-accounting network. We frame the explanatory logic (Tinbergen's four questions; ultimate vs. proximate causation; the adaptationist caveat), state the thesis (the sociometer as the master regulatory variable), and review seven mechanism families — costly signalling, prestige/dominance, reputation and indirect reciprocity, cheater detection and error management, coalitional psychology, mating-market value, and impression management — showing each as an operation on, or defence of, the value ledger. We close with the system-level synthesis, testable predictions, and the limits of the argument.

The central move of this paper is deflationary. It does not add a mechanism to the literature; it argues that the familiar mechanisms are subsystems of one regulator, and that naming the regulator explains why social behaviour has the shape it does — including the parts that look irrational from a within-lifetime, money-or-happiness point of view.

1. The question, and why proximate answers do not settle it

Ask why a person keeps a conversation going with someone whose offer they have no intention of taking, and the usual answer is "to be polite." That is true and explains nothing. Tinbergen (1963) separated four non-competing questions about any trait: its mechanism, its development, its phylogeny, and its *function* — the fitness problem it was built to solve. "Politeness" is a mechanism answer. The function question — what that disposition was selected *for* — is a different and deeper layer (Mayr's proximate/ultimate distinction; Scott-Phillips, Dickins & West, 2011), and conflating the two is the most common error in applied social psychology. This paper is an argument at the functional layer: the proximate feelings are real, but they are the readout of a system whose job is something the actor is not consciously tracking.

2. The explanatory frame

Behaviour is shaped by selection to have maximised **inclusive fitness in ancestral environments** (Hamilton, 1964), formalised in the gene-centred view (Williams, 1966; Dawkins, 1976). Two consequences structure everything below. First, organisms do not pursue fitness; they run the heuristics that, on average, produced it, which is why behaviour is coherent only once you ask what it was *for*, not what it currently optimises (Tooby & Cosmides, 1990). Second, sexual selection (Darwin, 1871) makes the honesty of quality the central adaptive problem of being chosen — the seed of §3.

Discipline is required. Not every regularity is an adaptation; some are by-products — the "spandrels" critique (Gould & Lewontin, 1979) — and adaptive narratives are cheap to generate and hard to test (Buller, 2005, is the rigorous internal critic). Accordingly the thesis below is offered as the best current integration of convergent evidence, not as proven from first principles, and §8 states what would falsify it.

3. The thesis: value to the group is the master variable

Human survival was never individual. Offspring are metabolically impossible for one adult to rear; provisioning, defence and teaching were collective (*Hrdy, 2009*, on cooperative breeding; Tomasello, 2009, on obligate shared intentionality). The decisive corollary: the group's aggregate appraisal of an individual's value *was* that individual's fitness, because the group determined access to food, mates, protection, and continued membership — and exclusion was lethal (*Boehm, 1999*; Kurzban & Leary, 2001). Selection therefore favoured a nervous system whose master regulatory task is not resource acquisition *per se* but standing: continuously estimate my value to those who can keep me alive, estimate others', and act to protect and raise my own.

The proximate instrument is the **sociometer** (Leary & Baumeister, 2000): self-esteem is not a private mood but a gauge tracking relational value, and its alarm is not metaphorical — social exclusion recruits the neural substrate of physical pain (Eisenberger, Lieberman & Williams, 2003), because in the ancestral world devaluation and injury were equally lethal. Belonging is thus a homeostatic system, not a preference (Baumeister & Leary, 1995).

> The unit of analysis is not the individual deciding about a proposition. It is two coupled processes running in every person at all times — advertise own value, appraise others' — because the network's output was survival.

This reframes "social dynamics" entirely. When the true quantity at stake (e.g. the merit of a proposition) is unassessable in the moment because of information asymmetry, the system substitutes a proxy it *can* assess: the standing and trustworthiness of the other party, and the consequence of agreement for one's own ledger entry. The mechanisms in §§4–10 are the subsystems that compute, signal, and defend that ledger.

| MASTER VARIABLE: my value to the group (sociometer) |

| PRODUCE value-signals | APPRAISE others' value | | · costly/honest signalling| · prestige attribution | | · prestige & competence | · reputation / gossip | | · impression management | · warmth × competence |

| DEFENCE (protect the estimate from exploitation): | | cheater detection · error management · coalitional | | vigilance · contempt / disqualification |

Figure 1. The value-ledger architecture. Status is the scoreboard; the listed mechanisms are its read/write/defence operations.

4. Honest signalling: why value cannot simply be claimed

When sender and receiver have non-aligned interests, communication is evolutionarily stable only if signals are costly enough to be hard to fake — the handicap principle (*Zahavi, 1975*), given a formal evolutionary-stability proof by Grafen (1990), and mirrored in economics as signalling theory (*Spence, 1973*). The rigorous treatment of when signals stay honest or collapse into deception is Searcy & Nowicki (2005); human ethnographic instances of value advertised through cost include conspicuous provisioning that benefits the signaller's standing more than their nutrition (Bliege Bird & Smith, 2005). The operative consequence: a claim of value is cheap and therefore discounted by design; demonstrated, costly, or third-party-verified value is what the appraisal machinery accepts. Cheap talk is not weak persuasion — it is the category of signal the receiver evolved to ignore.

5. Status as two routes: dominance and prestige

Rank — the coarse readout of group-conferred value — is obtained two ways (*Henrich & Gil-White, 2001*; Cheng, Tracy, Foulsham, Kingstone & Henrich, 2013): *dominance*, deference extracted through threat, and *prestige*, deference freely conferred on those whose competence benefits others. Prestige is the engine by which association transfers value: because humans preferentially copy and endorse the prestigious (Henrich, 2015), being seen to align with a high-value individual raises the aligner's own standing, and being associated with a discredited one lowers it. This is why, in persuasion, the appraised value of the source routinely overrides the content of the message: the listener is not only evaluating the claim, they are pricing what visible agreement does to their own ledger.

6. Reputation and indirect reciprocity: the distributed ledger

Cooperation at human scale is sustained by **indirect reciprocity**: benefit flows to those with good standing, and standing is maintained by third parties who were never themselves the beneficiaries (Trivers, 1971, on direct reciprocity; Nowak & Sigmund, 1998, 2005, on image scoring and indirect reciprocity). The bookkeeping technology is language itself: a substantial function of gossip is reputation accounting and the social brain that supports it (Dunbar, 1996). The ledger metaphor is therefore not loose — reputation is a literal, externally maintained record, which is why behaviour is sensitive to being observed and why even unobserved acts are rehearsed against an internalised audience.

7. The defensive subsystems: cheater detection and error management

A value-accounting animal is exploitable, so selection built an immune system for the ledger. Humans show content-specific facilitation on reasoning problems framed as social contracts — the cheater-detection result (Cosmides & Tooby, in Barkow, Cosmides & Tooby, 1992), though the modular interpretation remains debated. Its decision thresholds are not neutral: where the two error types had asymmetric ancestral costs, inference is biased toward the cheaper mistake — error management theory (Haselton & Buss, 2000). A persuasion attempt — a low-cost signal from a party whose payoff is decoupled from the receiver's — predictably trips this system, which is why a cooperative surface coexists with an adversarial appraisal underneath. "Humouring" is that configuration: the channel kept open while the defence runs to completion.

8. Coalitional and mating-market specialisations

The same accounting runs in domain-specialised forms. Coalitional psychology tracks alliance value and in-group/out-group boundaries because coalition membership was a determinant of survival (Boehm, 2012, on the band's reverse-dominance enforcement, which is the egalitarian flip side: over-display of dominance is *punished*, so value must often be signalled through restraint and benefit to others rather than through grabbing). In the mating market, the central problem is again deception about quality (Buss, 2003; Miller, 2000); partner-value is tracked continuously, and its withdrawal has a clear behavioural signature — contempt, which in dyadic research is the strongest single predictor of relationship dissolution (the Gottman programme). The folk notion of "testing" maps loosely onto adversarial probing of signal honesty; it should be treated as a hypothesis-generating gloss on partner-value assessment, not as an established construct.

9. The proximate interface: impression management and the two axes

At the surface, the system is operated through self-presentation: the self in interaction is a managed performance (Goffman, 1959, 1967), and politeness is the formal apparatus for protecting both parties' "face" (Brown & Levinson, 1987) — which is why the terminal social move is typically not hostility but the warm, low-cost disengagement that preserves every ledger entry, including the exiter's. Crucially, social perception cross-culturally resolves onto two dimensions — **warmth** (intent toward me) and **competence** (ability to act on it) (Fiske, Cuddy & Glick, 2007) — and warmth, the threat appraisal, is evaluated first and gates whether competence is even processed. Real-time dyadic signalling of this appraisal is measurable rather than merely interpretive (Pentland, 2008). The two axes are, in effect, the read-out format of the ledger: is this agent safe, and is this agent capable.

10. Synthesis

The mechanisms are not a list; they compose. The sociometer sets the regulated variable. Costly signalling solves the write problem (how value is credibly emitted). Prestige and the warmth–competence axes solve the read problem (how value is appraised and imported). Reputation and indirect reciprocity externalise the ledger across time and observers. Cheater detection and error management defend the estimate. Coalitional and mating systems are the same accounting in high-stakes domains. Impression management and face-work are the interface through which all of it is transacted in real time. Phenomena that look distinct — persuasion, deference, gossip, shaming, virtue display, conflict avoidance, the polite brush-off, contempt — are derivable as operations on a single quantity. The deflation is the contribution: one regulator, many surface forms.

11. Predictions and falsification

- **Source-before-content.** Manipulating appraised source value should shift acceptance more than equivalently manipulating argument quality, and the effect should grow as the proposition's merits become harder to verify independently.
- **Warmth gates competence.** Competence information should be down-weighted, not merely added to, when a prior warmth (threat) appraisal is negative — an interaction, not a main effect.

- **Observation sensitivity without belief in observation.** Behaviour should track reputational consequence even under conditions the actor explicitly believes are unobserved, because the ledger is internalised.
- **Restraint as signal where dominance is policed.** In contexts with strong reverse-dominance norms, value displays through self-limitation and conferred benefit should out-perform displays through resource capture.
- **Falsifiers.** The thesis is weakened if source-value and content effects are additive rather than interactive; if social-pain and physical-pain substrates dissociate cleanly under better designs; or if reputational sensitivity vanishes once belief-in-observation is fully removed. None of these is currently the consensus finding, but each is a clean test.

12. Limitations

This is a synthesis, not new data; its force is integrative coherence, not a novel result. Evolutionary-functional explanation is vulnerable to post-hoc storytelling (Gould & Lewontin, 1979; Buller, 2005), and several load-bearing components remain genuinely contested — the modularity of cheater detection, the status of group-level selection, and the empirical solidity of some signalling claims. The argument is strongest where multiple independent literatures converge (belonging/sociometer; prestige; reputation; warmth–competence) and should be held loosely where they do not. "Value to the group" is also a compression: it abstracts over kin vs. non-kin, short- vs. long-term, and domain-specific currencies that a fuller treatment must decompose.

13. Conclusion

The reason a person evaluates the messenger rather than the message, fears the chump role more than the loss, holds frame rather than argues, and exits politely rather than honestly, is not a collection of quirks. It is one system doing its job: regulating value in the only economy that, for almost all of human history, decided whether you survived. Name the regulator and the surface stops looking arbitrary.

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The Value Ledger · Closer Foundation working paper · theoretical synthesis, no new empirical data · attributions are to landmark sources and are load-bearing only where the cited literatures converge (see §12) · not for external distribution without review.

Citation gate, 2026-09-04

These eight were opened against a primary source this pass, because each carries a load-bearing claim in the text rather than a background one.

- **Boehm, C. (1999).** *Hierarchy in the Forest: The Evolution of Egalitarian Behavior.* Harvard University Press. `VERIFIED` The reverse-dominance hierarchy is his term and his thesis: egalitarianism is not the absence of hierarchy but a coalition of the rank and file deliberately dominating anyone who would dominate them.
- **Fiske, S. T., Cuddy, A. J. C., & Glick, P. (2007).** Universal dimensions of social cognition: warmth and competence. *Trends in Cognitive Sciences*, 11(2), 77–83. `VERIFIED` The two-axis model this paper leans on. Cite all three authors; a bare "Glick 2007" is incomplete.
- **Hrdy, S. B. (2009).** *Mothers and Others: The Evolutionary Origins of Mutual Understanding.* Harvard University Press. `VERIFIED` Cooperative breeding and alloparenting; supports the claim that a human infant is too costly for one adult to rear alone.
- **Leary, M. R., & Baumeister, R. F. (2000).** The nature and function of self-esteem: sociometer theory. *Advances in Experimental Social Psychology*, 32, 1–62. `VERIFIED` Self-esteem as a gauge of how much others value you as a partner.
- **Nowak, M. A., & Sigmund, K. (1998).** Evolution of indirect reciprocity by image scoring. *Nature*, 393(6685), 573–577. `VERIFIED` The formal result behind the reputation ledger: cooperation can pay without the same two individuals ever meeting again.
- **Spence, M. (1973).** Job market signaling. *Quarterly Journal of Economics*, 87(3), 355–374. `VERIFIED` The economic half of honest signalling. A signal separates types when it costs less to produce for those who actually have the quality.
- **Zahavi, A. (1975)** and **Grafen, A. (1990).** The handicap principle and its formal model. `CONTESTED` Penn & Számadó (*Biological Reviews*, 95(1), 267–290, online 2019) argue the strong version is an erroneous hypothesis that became a standard through misreadings of Grafen, and that signals need not be costly at equilibrium to stay reliable. Where this paper relies on costly signalling, read it as one mechanism rather than a law.

- **Henrich, J., & Gil-White, F. J. (2001).** The evolution of prestige: freely conferred deference as a mechanism for enhancing the benefits of cultural transmission. *Evolution and Human Behavior*, 22(3), 165–196. VERIFIED The dominance/prestige split: fear-based deference extracted, versus deference freely given to demonstrated skill.

Everything else cited in this paper is listed in its own reference section above with full details, and is LIKELY in the series' sense: real, canonical, and not re-opened this pass. The distinction is on the page rather than smoothed over, because the standing rule for this series is that a grade means something specific.
